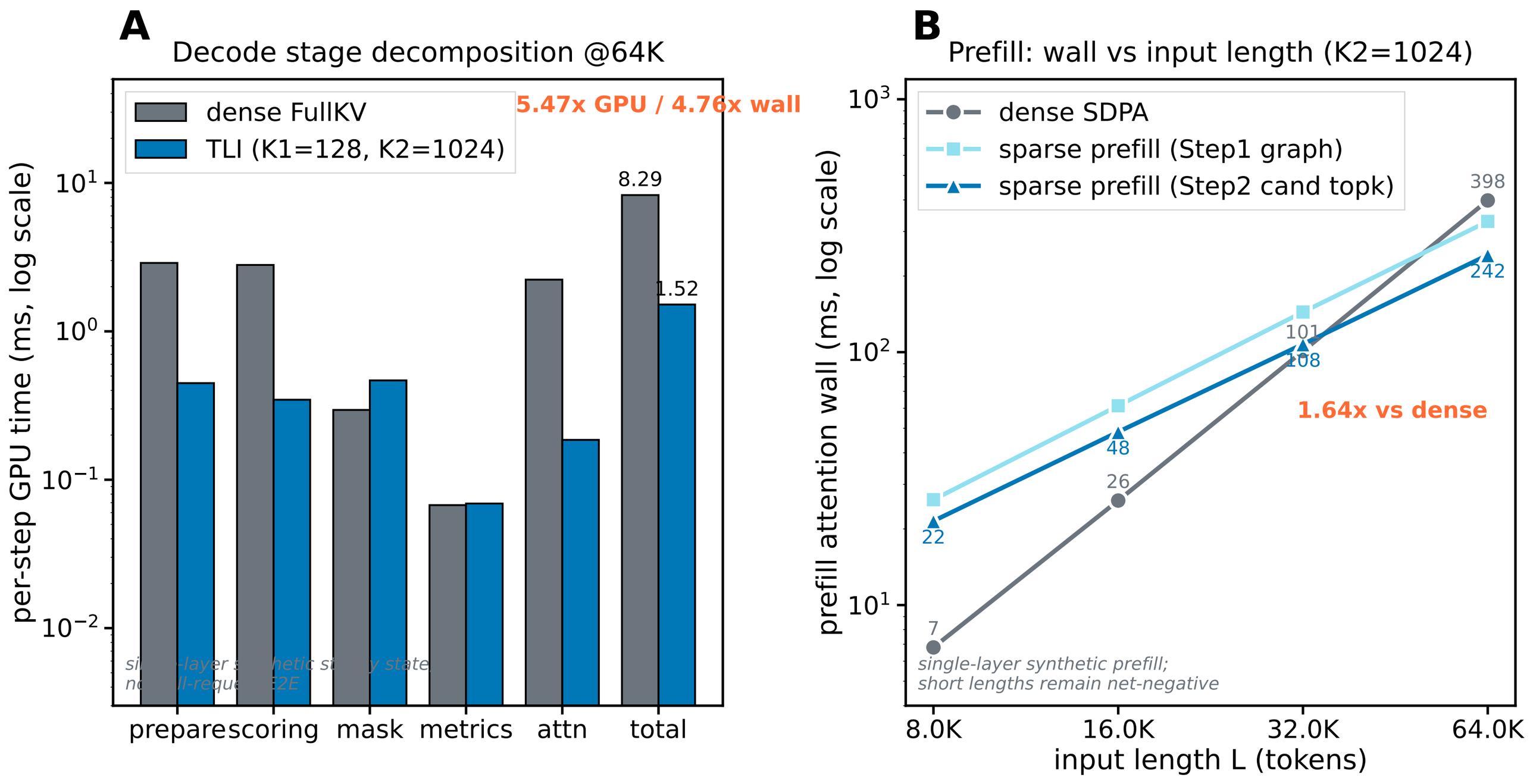
A
Decode stage decomposition @64K
dense FullKV
TLI (K1=128, K2=1024)
5.47x GPU / 4.76x wall
per-step GPU time (ms, log scale)
8.29
1.52
prepare
scoring
mask
metrics
attn
total
B
Prefill: wall vs input length (K2=1024)
dense SDPA
sparse prefill (Step1 graph)
sparse prefill (Step2 cand topk)
prefill attention wall (ms, log scale)
398
242
101
108
48
26
22
7
1.64x vs dense
single-layer synthetic prefill;
short lengths remain net-negative
8.0K
16.0K
32.0K
64.0K
input length L (tokens)